MIL-D-48016(MU)
7 November 1972
SUPERSEDING
See Section 6

MILITARY SPECIFICATION

DASHPOT ASSEMBLY ATOMIC WEAPON

1. SCOPE.

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of dashpot assembly used on M422 8-inch projectile.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

Fed. Test Method Std. No. 151: Metals; Test Methods
QQ-Z-325-Plating, Zinc, (Electrodeposited)
QQ-P-416-Plating, Cadmium, (Electrodeposited)

MILITARY

MIL-P-6871 -- Plating, Chromium

FSC 1110

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STANDARDS

MILITARY

MIL-STD-105 --- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1167 -- Ammunition Data Cards
MIL-STD-1168 -- Lot Numbering of Ammunition
MIL-STD-1169 -- Packaging, Packing and Marking for Shipments of Inert Ammunition Components

DRAWINGS

ORDNANCE CORPS

8787118 ------- Gasket
8787121 ------- Spacer
8787122 ------- Screw, Retaining
8787124 ------- Pin, Guide
8787251 ------- Reservoir
8787252 ------- Disc, Rupture
8794226 ------- Packing, Preformed
8794227 ------- Dashpot Assembly, Atomic Weapon
8794521 ------- Slug
8794593 ------- Disc, Screen
8794594 ------- Plug, Filling
8794595 ------- Disc, Orifice
8794597 ------- Detent Assembly
8799486 ------- Reservoir, Modified
8810177 ------- Gasket, Rubber Clad Steel
ET-8794227 ---- Equipment Tabulation

PUBLICATIONS

ASTM-E-8-66 --- Methods of Tension Testing of Metallic Materials.
ASTM-D88-56 --- Saybolt Viscosity

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The dashpot assembly shall comply with all requirements specified on the drawing (dwg) 8794227, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization. The contractor shall assign serial numbers as required by drawings. The contractor may select any number which complies with the drawing requirement. These numbers shall not be duplicated.

3.5 Silicone fluid. The silicone fluid viscosity shall comply with dwg requirements (see dwg 8794227) when tested as specified in 4.4.1. Fluid shall be discarded if exposed to temperatures below -55 degrees fahrenheit (^{o}F).

3.6 Leakage. The dashpot shall not show any evidence of leakage when tested as specified in 4.4.2.

3.7 Contractor's certication (see dwg 8787118, 8794595, and 8794226). The contractor shall certify in writing that:

a. The chemical composition and curing conditions of gasket;

b. The chemical composition of the packing, preformed;

c. The protective finish of the orifice disc comply with the applicable drawing requirements, and were taken from the lot as described herein. The certificate shall list the material tested, lot or batch number (as applicable), test performed, and the results obtained. The certification shall be signed by the contractor or a duly authorized agent, and presented to the Government inspector prior to or at the time of delivery of the lot.

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3.8 Workmanship.

3.8.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.8.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth, as required on the applicable drawing.

3.8.3 Burrs. Parts shall be free from burrs.

3.8.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.8.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted (see 6.4). When the contractor requires first article inspection, the contractor shall submit three (3) complete assemblies (see dwg 8794227) and one (1) set of parts for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table I. Tests need not be conducted in order specified in Table I.

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Table I

First Article Inspections and Tests

Inspection or Test	Dest	Non-Dest	Parts	Sub-Assy	Assy
1. Classification of characteristics (see 4.3.2.1)		X	X	X	X
2. Silicone fluid (see 3.5)	X		X		
3. Leakage (see 3.6)		X			X
4. Contractor's certification (see 3.7)		X	X		
5. Tensile (see 4.3.3.3)		X	X		
6. Salt spray and thickness of plating and adhesion of plating (see 4.3.3.4)	X		X		
7. Thickness of chrome plating and adherence test (see 4.3.3.5)		X	X		

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation. Lot formation shall be in accordance with MIL-STD-1168. Changes in log numbers shall be recorded on Ammunition Data Cards in accordance with MIL-STD-1167 and MIL-STD-1168.

4.3.1.1 Inspection lot. An inspection lot shall be the quantity of parts, subassemblies or assemblies from the same lot, which are submitted to the inspection for Government acceptance.

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4.3.2 Product inspection examination. Critical characteristics shall require 100 percent inspection of all items in the inspection lot. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table II and Table III), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Table II
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

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Table III

Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Spacer (see dwg 8787121) covering a detail of dwg 8794227.

Categories	Characteristics	Method of Inspection
Critical:	None defined	
Major:	None defined	
Minor		
201.	Diameter of hole, min.	Gage
202.	Outside diameter, max	Gage
203.	Thickness, max.	Gage
204.	Radius on hole proper	Visual
205.	Finish proper	Visual
206.	Workmanship (see 3.8)	Visual
207.	Plating, no bare spot	Visual

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4.3.2.1.2 Screw, retaining (see dwg 8787122) covering a detail of dwg 8794227.

Categories	Characteristics	Method of Inspection
Critical		
1.	Pitch diameter of thread, min.	Gage
2.	Major diameter of thread, min.	Gage
Major		
101.	Total length	Gage
Minor		
201.	Pitch diameter of thread, max.	Gage
202.	Major diameter of thread, max.	Gage
203.	Width of slot, min.	Gage
204.	Depth of slot, min.	Gage
205.	Inside diameter	Gage
206.	Finish proper	Visual
207.	Workmanship (see 3.8)	Visual/Manual

4.3.2.1.3 Pin, guide (see dwg 8787124) covering a detail of dwg 8794227.

Categories	Characteristics	Method of Inspection
Critical:	None defined	
Major:	None defined	
Minor		
201.	Small outside diameter	Gage
202.	Length of small outside diameter	Gage
203.	Large outside diameter	Gage
204.	Total length	Gage
205.	Specified corners broken	Visual
206.	Finish proper	Visual
207.	Workmanship (see 3.8)	Visual/Manual

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4.3.2.1.4 Gasket (see dwg 8787118) covering a detail of dwg 8794227.

Categories	Characteristics	Method of Inspection
Critical:	None defined	
Major		
101.	Contractor certification (see 3.7)	Visual
Minor		
201.	Thickness	Gage
202.	Inside diameter	Gage
203.	Outside diameter	Gage

4.3.2.1.5 Plug, filling (see dwg 8794594) covering a detail of dwg 8794227.

Categories	Characteristics	Method of Inspection
Critical:	None defined	
Major		
101.	Pitch diameter of thread, min.	Gage
102.	Major diameter of thread, min.	Gage
Minor		
201.	Pitch diameter of thread, max.	Gage
202.	Major diameter of thread, max.	Gage
203.	Small diameter	Gage
204.	Total length	Gage
205.	Length of small diameter	Gage
206.	Diameter of rear of thread	Gage
207.	Length of diameter to rear of thread	Gage
208.	Width of slot	Gage
209.	Depth of slot	Gage
210.	Finish proper	Visual
211.	Workmanship (see 3.8)	Visual/Manual

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4.3.2.1.6 Screen disc (see dwg 8794593) covering a detail of dwg 8794227.

Categories	Characteristics	Method of Inspection
Critical: None defined		
Major: None defined		
Minor		
201.	Outside diameter	Gage
202.	Type of mesh and width of opening correct	Visual

4.3.2.1.7 Disc, orifice (see dwg 8794595) covering a detail of dwg 8794227.

Categories	Characteristics	Method of Inspection
Critical		
1.	Diameter of hole	Gage
2.	Metal not defective; such as cracks, splits, etc.	Visual
3.	Workmanship (see 3.8)	Visual/Manual
4.	Hole diameter engraving complete, located correctly	Visual
Major		
101.	Thickness through bottom of cavity	Gage
102.	Contractor certification (see 3.7)	Visual
Minor		
201.	External diameter of disc	Gage
202.	Total width of disc	Gage
203.	Diameter of cavity	Gage
204.	Length of bevel	Gage
205.	Width of bevel	Gage
206.	Passivating treatment present	Visual
207.	Finish proper	Visual

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4.3.2.1.8 Slug (see dwg 8794521) covering a detail of dwg 8794227.

Categories	Characteristics	Method of Inspection
Critical: None defined		
Major		
101.	Diameter	Gage
102.	Finish proper	Test
Minor		
201.	Width of slot, max.	Gage
202.	Depth of slot, max.	Gage
203.	Radius on specified corner correct	Gage
204.	Perpendicularity of diameter with flat surface	Gage

4.3.2.1.9 Rupture disc (see dwg 8787252) covering a detail of dwg 8794227.

Categories	Characteristics	Method of Inspection
Critical		
1.	Thickness	Gage
Major		
101.	Diameter	Gage
102.	No cut, tear, fold or wrinkle	Visual
Minor: None defined		

4.3.2.1.10 Packing, preformed (see dwg 8794226) covering a detail of dwg 8794227.

Categories	Characteristics	Method of Inspection
Critical: None defined		
Major		
101.	Contractor certification (see 3.7)	Visual

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Minor
201. Workmanship (see 3.8) ..... Visual/Manual
202. No surface defects; cut, tear, voids, etc. ..... Visual

4.3.2.1.11 Detent assembly prior to plating and assembling slug (see dwg 8794597) covering a detail of dwg 8794227.

Categories	Characteristics	Method of Inspection
Critical: None defined		
Major: None defined		
Minor		
201.	Thickness between bottom of plug cavity and base surface	Gage
202.	Location of slug cavity	Gage
203.	Diameter of slug cavity and slug giving specified interference fit	Gage

4.3.2.1.12 Detent assembly prior to plating (see dwg 8794597) covering a detail of dwg 8794227.

Categories	Characteristics	Method of Inspection
Critical		
1.	Hardness test	Gage
2.	Large outside diameter, max.	Gage
3.	No crack seam, lap, fold or flake ...	Test (see 4.4.5)
4.	No slug loosely assembled as evidenced by shake or play (8794597 only)	Visual/Manual
Major: None defined		
Minor: None defined		

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4.3.2.1.13 Detent assembly (see 8794597) covering a detail of dwg 8794227.

Categories	Characteristics	Method of Inspection
Critical		
1.	Large outside diameter	Gage
2.	Group number stamp correct	Visual/Gage
3.	Distance from outside cylinder surface to large flat	Gage
4.	Location radius with outside diameter (functional)	Gage
5.	Location of slot with radius (functional)	Gage
6.	Location of large flat with radius (functional)	Gage
7.	Finish proper on large outside diameter	Visual
8.	Finish proper on large flat surface	Visual
9.	Finish proper on the radius of rear outside diameter	Visual
10.	Workmanship (see 3.8)	Visual
11.	Plating proper	Visual
Major:	None defined	
Minor		
201.	Radius at bottom of base cavity, max.	Gage
202.	Radius at outside diameter on base end correct	Gage
203.	Radius at intersection of shoulder and large flat correct	Gage
204.	Radius at front end correct	Gage
205.	Depth of drilled hole in side of body, min.	Gage
206.	Diameter of drilled hole in side of body, min.	Gage
207.	Concentricity of groove diameter with large outside diameter	Gage
208.	Diameter of groove	Gage
209.	Width of groove	Gage
210.	Radius at bottom of groove	Gage

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211.	Length from base end to shoulder	Gage
212.	Length from base end to bottom of front-end radius	Gage
213.	Distance from base end to slot, max.	Gage
214.	Depth of slot, min.	Gage
215.	Width of slot, min.	Gage
216.	Depth of base cavity, min.	Gage
217.	Diameter of base cavity, min.	Gage
218.	Location of groove from base end	Gage
219.	Radius at bottom of slot, max. (functional)	Gage
220.	Location of slot with outside diameter	Gage
221.	Finish proper on all surface, except large outside diameter, large flat surface end on the radius of the rear outside diameter	Visual
222.	Sharp corners removed as specified	Visual
223.	Outside diameter group number stamp correctly located	Visual
224.	Radius at intersection of shoulder and large flat does blend	Visual
225.	Angle at base cavity present	Visual
226.	Drilled hole in side of body not larger than width of slot	Visual
227.	Radius at mouth of base angle missing	Visual
228.	Serialization (see 3.4)	Visual

4.3.2.1.14 Reservoir prior to chromium plating (see dwg 8787251) covering a detail of dwg 8794227.

Categories	Characteristics	Method of Inspection
Critical		
1.	Large inside diameter	Gage
2.	Hardness	Gage
3.	No crack, seam, lap, fold or flake	Visual

Major: None defined

Minor: None defined

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4.3.2.1.15 Reservoir (see dwg 8787251) covering a detail of dwg 8799486.

Categories	Characteristics	Method of Inspection
Critical		
1.	Large inside diameter	Gage
2.	Group number stamp correct	Visual/Gage
3.	Finish proper on large inside diameter	Visual
4.	Workmanship (see 3.8)	Visual/Manual
5.	Plating proper	Visual
Major		
101.	Length from bottom of forward large radius to rear of flange surface, min.	Gage
102.	Diameter of shoulder under flange, max.	Gage
103.	Location of small hole through flange at flange slot with large inside diameter	Gage
104.	Location of cavity in flange with outside diameter below flange	Gage
105.	Location of large radius with outside diameter below flange	Gage
106.	Perpendicularity of flange shoulder with pitch diameter of outside threads	Gage
107.	Concentricity of pitch diameter of outside threads with outside diameter below flange	Gage
108.	Concentricity of outside diameter below flange with large inside diameter	Gage
109.	Position of hole through flange (.082) with .077 hole including position of (.082) hole with slot	Gage
110.	Angular position of (.077) hole with respect to centerline with (3.243) radius	Gage
Minor		
201.	Diameter of shoulder under flange, min.	Gage
202.	Length from bottom of forward large radius to rear of flange surface, max.	Gage
203.	Pitch diameter of outside threads	Gage

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204.	Major diameter of outside threads	Gage
205.	Pitch diameter of side hole threads	Gage
206.	Minor diameter of side hole threads, max.	Gage
207.	Pitch diameter of base end hole threads	Gage
208.	Minor diameter of base end hole threads, max.	Gage
209.	Depth of side hole threads, min.	Gage
210.	Depth of base end hole threads, min.	Gage
211.	Depth of large cavity, min.	Gage
212.	Profile of front end radius, min.	Gage
213.	Diameter of small hole at flange slot	Gage
214.	Diameter of small hole in flange	Gage
215.	Width of slot on flange, min.	Gage
216.	Location of side threaded hole	Gage
217.	Diameter of boss at bottom of large cavity, max.	Gage
218.	Depth of large cavity to top surface of base, min.	Gage
219.	Depth of base end threaded cavity (see dwg 8787251 and 8799486, as applicable)	Gage
220.	Depth of side threaded cavity, min.	Gage
221.	Diameter of center hole, min.	Gage
222.	Diameter of hole below side threaded cavity	Gage
223.	Length from shoulder under flange to flange	Gage
224.	Length from base end to outside thread, max.	Gage
225.	Length of outside thread undercut, max.	Gage
226.	Total length	Gage
227.	Diameter of flange	Gage
228.	Distance from center of body to flange slot	Gage
229.	Radius at bottom of base end threaded hole, max.	Gage
230.	Radius between rear surface of flange and diameter under flange, max.	Gage
231.	Position of groove under flange shoulder	Gage
232.	Width of groove under flange shoulder	Gage
233.	Depth of groove under flange	Gage
234.	Large radius at base end present	Visual
235.	Radius between bottom of inside large cavity and undercut groove present	Visual

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236. Radius between bottom of inside large cavity and boss present ..... Visual
237. Radius at forward edge of boss present ..... Visual
238. Sharp corners removed as specified ..... Visual
239. Large inside diameter group number stamp present or located correctly ..... Visual
240. Undercut at bottom of large inside cavity present ..... Visual
241. Workmanship (see 3.8) ..... Visual
242. Finish proper on all surfaces except inside diameter ..... Visual
243. Serialization (see 3.4) ..... Visual

4.3.2.1.16 Reservoir, modified (see dwg 8799486) covering a detail of dwg 8794227.

Categories	Characteristics	Method of Inspection
Critical:	None defined	
Major		
101.	Diameter of thread clearance, max.	Gage
102.	Depth of thread including thread clearance, max.	Gage
Minor:	None defined	

4.3.2.1.17 Dashpot assembly, atomic weapon prior to assembling any components in reservoir (see dwg 8794227).

Categories	Characteristics	Method of Inspection
Critical:	None defined	
Major		
101.	No foreign matter in detent cavity	Visual
102.	No foreign matter in retaining screw cavity ...	Visual
103.	No foreign matter in filling plug cavity	Visual
104.	No foreign matter on detent including O-ring (except for silicone fluid)	Visual
105.	O-ring adequately seated on detent	Visual

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Minor: None defined

4.3.2.1.18 Dashpot assembly, atomic weapon prior to assembling retaining screw, spacer, gasket, disc, screen, orifice disc, rupture disc, O-ring and filling plug (see dwg 8794227).

Categories	Characteristics	Method of Inspection
Critical: None defined		
Major		
101.	Filling pin engaging filling pin slot in detent	Visual/Manual
102.	Hole in orifice disc not obstructed (including proper size of hole depending on type specified on applicable drawing)	Visual/Gage
Minor: None defined		

4.3.2.1.19 Dashpot assembly, atomic weapon (see dwg 8799487).

Categories	Characteristics	Method of Inspection
Critical		
1.	Group number stamps on detent and reservoir in the same grouping	Visual
Major		
101.	Retaining screw tight or adequately torqued	Gage
102.	Guide pin fully seated	Visual/Manual
103.	Filling plug tight	Visual/Manual
104.	O-ring adequately assembled in groove under flange	Visual/Manual
105.	No lacquer on unpermitted surfaces	Visual
Minor		
201.	Height of spring pin	Gage
202.	Marking correct, complete or identifiable	Visual
203.	Serialization (see 3.4)	Visual

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4.3.2.1.20 Gasket, rubber clad steel prior to rubber coating (see dwg 8810177) covering a detail of dwg 8794227.

Categories	Characteristics	Method of Inspection
Critical: None defined		
Major: None defined		
Minor		
201.	Inside diameter	Gage
202.	Outside diameter	Gage
203.	Thickness	Gage
204.	Concentricity of inside diameter with outside diameter	Gage
205.	Workmanship (see 3.8)	Visual

4.3.2.1.21 Gasket, rubber clad steel (see dwg 8810177) covering a detail of dwg 8794227.

Categories	Characteristics	Method of Inspection
Critical		
1.	No presence of rubber on outside surfaces (presence of rubber on inside diameter permissible, but not required) (a five (5) power glass or five (5) power binocular microscope shall be used)	Visual
Major		
101.	Thickness	Visual
Minor		
201.	Inside diameter, min.	Gage
202.	Outside diameter, max.	Gage

4.3.2.1.22 Preparation for delivery (see Section 5).

Categories	Characteristics	Method of Inspection
Critical: None defined		
Major		
	Preservation, packaging, packing and marking in accordance with MIL-STD-1169	Visual/Manual
Minor: None defined		

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4.3.3 Testing. All tests shall be performed. Tests need not be performed in the order specified.

Test	Test Classification
a. Silicone fluid	Major
b. Leakage	Major
c. Tensile	Major
d. Salt spray, thickness of plating and adhesion of plating	Major
e. Thickness and adhesion of chrome plating	Major

Of these tests, d. is considered destructive, and samples so tested shall not be returned to the lot. Destructive samples are over and above the deliverable quantity of items required by the contract.

4.3.3.1 Silicone fluid (see 3.5). Five samples, each one sufficient to conduct the test, shall be selected from each lot of silicone fluid and tested in accordance with 4.4.1. If any sample fails to comply with 3.5, the lot of silicone fluid shall be rejected. If the fluid is used for more than 12 months since the last acceptance test, it shall be retested prior to use. This policy shall be followed every 12 months until the fluid is used.

4.3.3.2 Leakage (see 3.6). Each dashpot assembly shall be tested in accordance with 4.4.2. If any dashpot assembly fails to comply with 3.6, the dashpot shall be rejected and removed from the lot.

4.3.3.3 Tensile (see dwgs 8787251 and 8794597). Five samples of the material heat treated simultaneously with the parts shall be prepared and tested in accordance with 4.4.3. If any specimen fails to comply with drawing requirements, the heat treated lot shall be rejected.

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4.3.3.4 Salt spray, thickness of plating, and adhesion of plating (see dwgs 8787121, 8787124 and 8794594). Samples shall be selected in accordance with QQ-Z-325 and QQ-P-416 as applicable and tested in accordance with 4.4.4. If any sample fails to comply with QQ-Z-325 and QQ-P-416, as applicable, the lot of plated parts shall be rejected.

4.3.3.5 Thickness of chrome plating and adherence test (see dwg 8787251 and 8794597). Five samples shall be plated simultaneously with the parts and tested in accordance with 4.4.5. If any sample fails to comply with the requirements of MIL-P-6871, the lot of plated parts shall be rejected.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Critical and major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL's). EL's for assemblies are listed on the Equipment Tabulation, ET-8794227. When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).

b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on EL's. The acceptance inspection equipment operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances, less the accuracy tolerances of the test equipment used.

4.4.1 Silicone fluid (see 3.5). The viscosity of the silicone fluid shall be tested in accordance with ASTM-D88-56.

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4.4.2 Leakage (see 3.6). The dashpot assembly shall be supported in an Ordnance Approved Test fixture so that the detent is in an upright position. A 25-pound (lb) ± 5% lb load shall be placed on top of the detent for a period of not less than five minutes. While under load, if the load decreases, reject the assembly. Remove the load and examine the dashpot assembly for fluid leaks (see 3.5). The rejected dashpot may be rebuilt and retested for acceptance.

4.4.3 Tensile (see 4.3.3.3). The tensile strength shall be tested in accordance with ASTM-E-8-66.

4.4.4 Salt spray, thickness of plating and adhesion of plating (see 4.3.3.4). Samples shall be tested in accordance with QQ-Z-325 and QQ-P-416, as applicable.

4.4.5 Thickness of chrome plating and adherence test (see 4.3.3.5). Samples shall be tested in accordance with MIL-P-6871.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The dashpot assembly shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.1.2 Level C. The dashpot assembly shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.2 Packing.

5.2.1 Levels A and C. The dashpot assembly shall be prepacked in accordance with the applicable instructions contained in MIL-STD-1169(MU).

5.3 Marking. Marking shall be in accordance with MIL-STD-1169(MU).

5.4 Data cards. Data cards shall be prepared in accordance with MIL-STD-1167.

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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number and date of this specification.

b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1 and 5.4). Data specified in 4.3.4.1 and 5.4 will be delivered as identified on a numbered DD Form 1664, when specified on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commanding Officer Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SMUPA-QA-N-I.

6.4 Instructions for first article inspection (see 4.2.1). Parts may be used in subassemblies if the examination or tests (see 4.2.2, Table I) are verified by a representative designated by the Government. Subassemblies may be used in final assemblies if the examination or tests (see 4.2.2, Table I) are verified by a representative designated by the Government.

6.5 Supersession data. This specification supersedes the requirements of Purchase Description PA-PD-1056 (Rev 1), dated 9 February 1962.

Custodian:
Army - MU

Preparing activity:
Army - MU

Project No. 1110-A038

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-D-48016(MU) Dashpot Assembly Atomic Weapon

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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DEPARTMENT OF THE ARMY

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OFFICIAL BUSINESS

Commanding Officer
Picatinny Arsenal
ATTN: SMUPA-NR
Dover, New Jersey 07801

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